01 / PROBLEM AND CONTRIBUTION

FIFO checking with DPI-C & assertions

Advanced Digital Design & Verification · Lab 4 | Fall 2026 | Collaborative lab

RESULT AT A GLANCE

Checked 80 writes and 80 reads with zero errors in the normal DPI run; an intentional fault injected at 140 ns was detected at 145 ns, a 5 ns interval in that simulation.

The engineering question

How can a verification environment show that it both accepts correct behavior and detects a deliberate error?

Situation

A checker must recognize correct transactions and also demonstrate that it catches errors. This project pairs a C queue model with SystemVerilog assertions and an intentional fault experiment.

Task

Compare accepted FIFO operations through DPI-C, check stream-control properties, and test the routing assertion with a deliberate bad control signal.

My contribution

Collaborative verification work with Akshith Alwala, covering the C reference model, DPI checking, assertions, and simulation-based fault testing.

Tools and concepts

SystemVerilog Assertions / DPI-C / C / Reference models / Fault injection

02 / DESIGN AND METHOD

From the task to implementation

01 / Implementation step

Used a 16-entry C queue with head, tail, and count as a behavioral reference model, accessed from SystemVerilog through DPI-C. Compare accepted operations instead of assuming the two clock domains observe occupancy simultaneously.

02 / Implementation step

Checked full/empty protection, wraparound, and overlapping traffic. For the alternating stream, added 11 assertions for routing, read selection, state transitions, parity, and queue depth.

03 / Implementation step

Injected an incorrect even-write control to confirm the routing assertion detects the fault; kept this intentional failure separate from normal regressions.

Independent behavioral checking

The C reference queue tracks head, tail, and count through DPI-C calls. Comparing accepted operations avoids treating independently clocked occupancy observations as though they were simultaneous.

Architecture / workflow

Conceptual summary of the documented workflow; not a simulation result.

01	Accepted operations
02	C golden queue
03	DPI checker
04	SVA / injected faults

03 / RESULTS AND ENGINEERING PERSPECTIVE

Evidence, interpretation, and next steps

Writes / reads	80 / 80
Checker errors	0
Stream assertions	11
Fault-detection interval	5 ns

Reported result

The reported normal DPI run completed 80 writes and 80 correct reads with zero checker errors. In the intentional faulty run, an injected routing fault at 140 ns triggered assertion A1 at 145 ns.

Testing the checker

A deliberate routing-control fault exercises assertion A1. The 5 ns interval is the observed time from injection to detection in that test; it is not a general latency bound for all faults or clock configurations.

Scope and limitations

This is simulation-based verification. The report discusses formal checks as a possibility but does not report a formal-tool proof.

Possible next steps - proposed, not completed results

- Package the normal and intentional-failure regressions with clearly different expected outcomes.
- Add more fault classes and document assertion coverage before claiming broader verification completeness.

Source material: ADDV Lab 4 Parts 1-4 updated report, sections 2 and 4.

Prepared from the supplied reports, project files, and verified portfolio copy. Figures retain their original content; explanatory diagrams are labelled.